MINI PROJECT 2 BONUS

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Comparative Performance Analysis of RNNs (GRU/LSTM) vs Transformer for Time-Series Classification

1. Introduction

This study compares the performance of Recurrent Neural Networks (RNNs), specifically GRU/LSTM, with Transformer-based models for multivariate time-series classification. The task involves predicting class labels from sequential sensor data.

2. Experimental Setup

- Input: Time-series sequences (length = 100, multiple features)
- Classes: 5 categories
- Models used:
  - GRU-based RNN
  - Transformer with self-attention
- Evaluation metrics:
  - Accuracy
  - Loss
  - Confusion Matrix

3. Transformer Performance (Final Results)

From training logs:

- Training Accuracy: ~90.5%
- Validation Accuracy: ~88.3%
- Validation Loss: ~0.298

Key Observation:

- The model shows strong learning capability
- Learning rate scheduling helped improve convergence:
  - Reduced from 3.125e-05 → 1.56e-05
- Small gap between training and validation accuracy → good generalization

4. GRU / RNN Performance (Observed Earlier)

- Training Accuracy: ~86–88%
- Validation Accuracy: ~85–87%
- More stable training behavior
- Lower sensitivity to hyperparameters

5. Performance Comparison

Metric	GRU (RNN)	Transformer
Training Accuracy	~86–88%	~90.5%
Validation Accuracy	~85–87%	~88.3%
Validation Loss	Higher	Lower (~0.29)
Learning Capability	Good	Excellent
Generalization	Stable	Improved with tuning

6. Key Observations

6.1 Accuracy Improvement

The Transformer model outperformed GRU by approximately 2–3% in validation accuracy, indicating better feature learning and representation.

6.2 Effect of Learning Rate Scheduling

The use of ReduceLROnPlateau significantly improved performance:

- Allowed finer convergence
- Prevented overshooting minima
- Resulted in better validation accuracy

6.3 Model Learning Behavior

GRU:

- Learns sequentially step-by-step
- Strong temporal bias
- Performs well on moderate datasets

Transformer:

- Uses self-attention to analyze entire sequence
- Captures long-range dependencies
- Learns complex feature relationships

6.4 Generalization Ability

- Transformer achieved higher accuracy with controlled overfitting
- Gap between training and validation accuracy is small:
    - Indicates good generalization

7. Confusion Matrix Insights

- Majority of predictions lie along the diagonal
- Misclassifications occur between similar classes
- Transformer shows improved class separation compared to GRU

8. Why Transformer Performed Better (Final Model)

The improved Transformer achieved better performance due to:

- Increased model capacity (more heads and layers)
- Proper dimensional alignment
- Learning rate scheduling
- Better regularization (dropout)
- Enhanced feature learning via attention

9. Conclusion

The Transformer model outperformed the GRU-based RNN in this task, achieving higher accuracy and better generalization.

Final Insight:

- RNNs are efficient and stable for sequence modeling
- Transformers, when properly tuned, can surpass RNNs by capturing global dependencies and complex feature interactions

10. Final Statement (Use in Viva)

"Although GRU models are effective for sequential data, the Transformer achieved better performance due to its self-attention mechanism, which captures both local and global dependencies more efficiently."

11. Future Scope

- Hybrid models (CNN + Transformer)
- Larger datasets for further improvement
- Real-time implementation
- Attention visualization for interpretability